Energy Wave

CONFIDENTIAL — DRAFT — FOR DISCUSSION · BOUNDS ARE MEASURED AT COMMISSIONING, NOT MODELLED HERE

The platform is the asset whatever the gates find. Its error budget is a table of measured upper bounds.

The September prospectus commits the company to publish the measurement platform's design and its error budget as a methods paper before any result is claimed. The budget assigns to every known way of manufacturing a picowatt-level signal a *measured* upper bound on the spurious output it could produce in the platform as built. The artifact floor is the quadrature sum of those bounds; the platform is not commissioned until the floor is at least a factor of ten below the effect it is built to test. This page is that table with the method for each row and the bounds left to be measured.

1 · THE COMMISSIONING CRITERION

> **Artifact floor** = $\sqrt{(\Sigma\ \text{bound}_i^2)}$ over every row of §2, each bound measured on dead devices, twins and fixture-only runs in the platform as built. **Commissioned** when floor ≤ (reported effect) ÷ 10. For thread B the reported effect is 1.4 pW (Moddel et al. 2021, one laboratory, unreplicated), so the plan proposes a floor at or below 0.14 pW — a commissioning target set by this plan at one-tenth of the claimed maximum, not an established requirement; the thread C lead and the advisory board ratify it at commissioning. For thread A the effect is the smallest conditioning-dependent change the pre-registration declares meaningful. **[Needs Input from Ian or Chip: the thread A effect size the plan must resolve]**

2 · THE ERROR-BUDGET TABLE

Artifact source	How the bound is measured	Instrument / fixture	Measured bound
Thermal gradient (Seebeck)	Map ΔT across the device site with distributed sensing; sweep ±ΔT deliberately; record output of dead devices and no-cavity twins versus ΔT; bound = output at the residual gradient after stabilization.	Multi-stage actively stabilized isothermal enclosure; distributed temperature sensing	[to be measured] W at residual ΔT = [ ] mK
Photo-effects	Measure light leakage in the closed enclosure; sweep controlled illumination; bound = output at measured leakage on mirror-only twins.	Light-tight enclosure; calibrated photodiode; controlled source	[to be measured]
Electromagnetic pickup and rectification	Inject known RF at swept frequency and power; record device response; measure residual ambient field inside shielding; bound = response at residual ambient.	Nested conductive and high-permeability shielding; filtered feedthroughs; guarded triaxial wiring; spectrum monitor	[to be measured]

Artifact source	How the bound is measured	Instrument / fixture	Measured bound
Contact potential and electrochemistry	Humidity-controlled and vacuum runs; discharge-and-recovery tests; long-term drift on twins; bound = steady residual after discharge.	Humidity control; vacuum option; electrometer	[to be measured]
Stored charge and dielectric relaxation	Extended discharge to ground before every run; relaxation-time fits on spacer-only twins; bound = residual current after the declared discharge interval.	Electrometer; timed grounding fixture	[to be measured]
Triboelectric and piezoelectric	Correlate output with accelerometer spectrum; isolation on and off; bound = output at the residual vibration spectrum.	Passive isolation; accelerometers	[to be measured]
Instrument offsets and bias current	Interleave dead-device, open-circuit and shorted runs with every measurement; characterize offset drift over the run length; bound = drift-corrected residual.	Electrometer-class instrumentation with characterized bias current	[to be measured]
Leakage paths	Fixture-only runs; guarded fixtures; bound = fixture output with no device.	Guarded fixture	[to be measured]
Probe phase skew (power measurements)	Full-chain calibration; probe swap; frequency sweep against the deskew model; bound = residual real-power error at the highest frequency used.	Sampled v(t), i(t) acquisition; deskew fixture	[to be measured]
Selection and analysis bias	Not a measured bound: pre-registered device matrix, blinding keys held outside the team, fixed stopping rules, all devices reported.	Registry deposit; data pipeline with hashing at acquisition	procedural
Artifact floor	**√Σ bound²** = [to be computed at commissioning] — commissioned when ≤ 0.14 pW for thread B (proposed target).		

3 · COMMISSIONING SEQUENCE

1. Build and close the enclosure; measure shielding effectiveness versus frequency and the residual ambient field.
2. Map the thermal field at the device site over 24 h; set the stability and residual-gradient targets in the pre-registration.
3. Characterize the electrometer chain on open, short and known-null loads; record bias current and offset drift.
4. Run dead devices, fixture-only and each twin class through the full measurement sequence; fill every bound in §2 from those runs.
5. Compute the floor; compare with the criterion; publish the methods paper with the table; only then measure a live device.

4 · CALORIMETRY — COMMISSIONED IN GATE 1, DECISIVE IN GATE 2

A differential calorimeter with a reference cell alongside the device cell, so that heat flow into the device can be measured while output is drawn. Its resolution sets what Gate 2 can say: if output is too small to resolve

calorimetrically, Gate 2 requires the device count and integration time needed to resolve it before any conclusion is drawn. Resolution target **[Needs Input from Ian or Chip: the calorimetric resolution the program will fund]**.

5 · SIGN-OFF

Role	Responsibility	Named
Measurement and instrumentation lead (thread C)	Owns the table, the commissioning runs and the methods paper.	**[Needs Input from Ian or Chip: name, or “to hire — a Tranche A condition”]**
Principal investigator	Signs the pre-registration; certifies the floor before live devices.	**[Needs Input from Ian or Chip: name, or “to hire”]**
External scientific advisory board	Reviews the table before publication; holds the veto over performance statements.	To recruit — at least one declared critic

Sources: Technical Prospectus v1 §§6.3–6.4, 7.1, Appendix B (M-05); Physical Architecture & Baseline Model v1.3 §3 (M-14); Investor Overview slides 10, 12 (M-04); Research Program and Gates page; Thread A plan §5.